

Edward Chen

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Education

Arizona State University

August 2023 - May 2027

Major: BS Data Science, Minor in Economics (GPA: 3.7)

Coursework: Data Structures, Operating Systems, Computer Vision, Deep Learning, Machine Learning, Linear Algebra

Activities: Scholars of Finance, Association of Computing Machinery

Experience

Age of Learning | Data Engineer Intern

June 2026 - August 2026

- Built an internal AI agent platform with custom tools, persistent context, and subagent execution, enabling secure access to production systems and automated GitLab merge-request reviews.
- Deployed monitoring for reports backed by a 314M-row production table, tracing data failures across 4,000+ data models to the responsible pipeline or code change.
- Extended a personalized activity-ranking system while preserving curriculum requirements, improving learner continuation after lessons by 20% in an A/B test.

[mymelo.org](#) | Co-Founder, Software Engineer

April 2025 - June 2026

- Co-founded Melo as a winning project at HackTech and scaled it into a patient-and-provider platform for continuous skin monitoring and risk-based triage.
- Trained EfficientNet-Bo on 30K dermoscopy images with transfer learning and class-weighted augmentation, achieving 95% recall for melanoma triage on a held-out test split.
- Engineered AWS S3 + Lambda pipeline to compress 3-5MB images to 300KB WebP, reducing per-image disk footprint by 94% while preserving originals in S3 for compliance. Built mobile platform end-to-end.

M.Y Intellectual Property | Software Engineer Intern

May 2024 - August 2024

- Engineered a unified patent search API aggregating USPTO, CNIPA, and EPO databases into a single query interface, reducing cross-jurisdictional research from 3 workflows to 1 and cutting attorney lookup time by 60%.
- Built citation network parser to extract patent family relationships across 50K+ international filings, automating competitor portfolio analysis for 15+ client engagements.

Projects

KV-Cache Compression for Transformer Inference | [local-turboquant](#) | PyTorch, Modal, transformers, CUDA

- Shipped a pip-installable library with a one-line activate() API, wrapper, and CLI exposing per-generation savings, throughput, and free VRAM; benchmarked on H100s via Modal.
- Fused Triton attention kernel that runs directly on bit-packed 3-bit KV indices without decompressing to FP16, achieving 74% KV cache memory reduction at near-lossless quality

Modified Transformer Architecture | [smarternano](#) | PyTorch, CUDA, OpenRouter, transformers

- Re-engineered the nanochat architecture with custom layer depth and weight initialization, re-trained on Nvidia's Nemotron Nano dataset, publicly recognized by Andrej Karpathy.
- Built a synthetic data generation pipeline via OpenRouter to align the model's personality, achieving coherence comparable to 560M parameter models.

AI-Native App | [murmur](#) | ONNX, Flutter, AI Engineering

- Built and released a mobile app that converts any PDF into human-quality audio entirely on-device, achieving 3-second load-to-playback with zero server dependencies. Deployed Kokoro-82M transformer TTS on Android via ONNX Runtime, forking the runtime to isolate inference on a background thread for performance.
- Built an NLP preprocessing pipeline by cross-compiling espeak-ng and writing Dart FFI bindings for IPA phonemization, implementing token-aware batch splitting against the model's tokenizer vocab, and caching per-sentence phonemes in SQLite to eliminate redundant compute at inference time.

Skills

Languages: C++, Python, Java, CUDA

Technologies: AWS, Docker, Flutter, React, Next.js, Django, Flask, Supabase, Git, MongoDB, Snowflake, SQLite

AI/ML: PyTorch, FastAPI, ONNX Runtime, RL, vLLM, SFT, LoRA, FAISS, Synthetic Data, Transformers, JAX

Contributions: Meta PyTorch